Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the modulus and principal argument of $z = -2\sqrt{3} + 2i$, and express z in polar form $r(\cos\theta + i\sin\theta)$.

Step 1: Compute modulus r: $r = \sqrt{(-2\sqrt{3})^2 + 2^2} = \sqrt{12+4} = \sqrt{16} = 4$. Step 2: Determine argument θ: Since $a = -2\sqrt{3} < 0$ and $b = 2 > 0$, z lies in Quadrant II. Reference angle: $\tan\alpha = \left|\frac{2}{-2\sqrt{3}}\right| = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \frac{\pi}{6}$ (or 30°). Quadrant II angle: $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ (or 150°). Step 3: Write polar form: $z = \mathbf{4}\left(\cos\frac{\mathbf{5\pi}}{\mathbf{6}} + \mathbf{i}\sin\frac{\mathbf{5\pi}}{\mathbf{6}}\right)$.

2 **[Foundational]** • **Target: Powers of i**

Evaluate each power of i, using $i^2 = -1$ and the fact that powers of i repeat every 4:

a $i^3 = -i$ **b** $i^4 = 1$ **c** $i^7 = -i$ **d** $i^{22} = -1$

(a) $= -i$; (b) $= 1$; (c) $= -i$; (d) $= -1$

3 **[Foundational]** • **Target: Powers of i**

Evaluate i^{101}.

$101 = 4(25) + 1$, so $i^{101} = (i^4)^{25} \cdot i^1 = 1 \cdot i = i$.

4 **[Foundational]** • **Target: Addition and subtraction**

Let $z_1 = 3 + 4i$ and $z_2 = -2 + 5i$. Find $z_1 + z_2$.

$= 1 + 9i$

5 **[Foundational]** • **Target: Addition and subtraction**

Let $z_1 = 2 - 3i$ and $z_2 = -1 - 4i$. Find $z_1 + z_2$.

$= 1 - 7i$

6 **[Foundational]** • **Target: Addition and subtraction**

Find $(7 + 2i) - (3 - 5i)$.

$= 4 + 7i$

7 **[Foundational]** • **Target: Addition and subtraction**

Find $(4 - i) - (-2 + 6i)$.

$= 6 - 7i$

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Multiplication**

Multiply $(2+3i)(1-4i)$.

$(2)(1)-(3)(-4)+[(2)(-4)+(3)(1)]i = 14-5i$

9 **[Intermediate]** • **Target: Multiplication**

Multiply $(-1+2i)(3+i)$.

$(-1)(3)-(2)(1)+[(-1)(1)+(2)(3)]i = -5+5i$

10 **[Intermediate]** • **Target: Multiplication**

Multiply $(5-2i)(5+2i)$, and explain why the result has no imaginary part.

$= 25-(-4)+0i = 29$. It is a product of conjugates $z\bar{z} = |z|^2$ (Theorem 1.1), which is always real.

11 **[Intermediate]** • **Target: Conjugate and modulus**

For $z = 6-8i$, find $\bar{z}$ and $|z|$.

$\bar{z} = 6+8i$, and $|z| = \sqrt{6^2+8^2} = \sqrt{100} = 10$.

12 **[Intermediate]** • **Target: Conjugate and modulus**

For $z = -5+12i$, find $\bar{z}$ and $|z|$.

$\bar{z} = -5-12i$, and $|z| = \sqrt{5^2+12^2} = \sqrt{169} = 13$.

13 **[Intermediate]** • **Target: Conjugate and modulus**

Show that $z+\bar{z}$ is always real for $z = a+bi$.

$z+\bar{z} = (a+bi)+(a-bi) = 2a$, which is real since a is real.

14 **[Intermediate]** • **Target: Division**

Simplify $\frac{3+4i}{1-2i}$ by multiplying top and bottom by the conjugate of the denominator.

Multiply by $\frac{1+2i}{1+2i}$: numerator $= (3)(1)-(4)(2)+[(3)(2)+(4)(1)]i = -5+10i$; denominator $= 1^2+2^2 = 5$. So the quotient is $-1+2i$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Division • Cross-Lesson: 6.01 Limits

Simplify $\frac{1+i}{1-i}$.

Multiply by $\frac{1+i}{1+i}$: numerator $= (1)(1) - (1)(1) + [(1)(1) + (1)(1)]i = 0 + 2i$; denominator $= 1^2 + 1^2 = 2$. So the quotient is i.

16 [Challenging] • Target: Division • Cross-Lesson: 2.01 Matrices & Vectors

Simplify $\frac{7-i}{3+2i}$.

Multiply by $\frac{3-2i}{3-2i}$: numerator $= (7)(3) - (-1)(-2) + [(7)(-2) + (-1)(3)]i = 19 - 17i$; denominator $= 3^2 + 2^2 = 13$. So the quotient is $\frac{19}{13} - \frac{17}{13}i$.

17 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 3.01 Sequences & Series

Solve $x^2 + 9 = 0$.

$x^2 = -9 \Rightarrow x = \pm 3i$.

18 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 4.01 Advanced Functions & Equations

Solve $x^2 - 6x + 13 = 0$ using the quadratic formula.

Discriminant $= 36 - 52 = -16$, so $\sqrt{-16} = 4i$. $x = \frac{6 \pm 4i}{2} = 3 \pm 2i$.

19 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve $2x^2 + 2x + 5 = 0$ using the quadratic formula.

Discriminant $= 4 - 40 = -36$, so $\sqrt{-36} = 6i$. $x = \frac{-2 \pm 6i}{4} = -\frac{1}{2} \pm \frac{3}{2}i$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Limits

Simplify $\frac{3+4i}{1-2i}$ by multiplying top and bottom by the conjugate of the denominator.

Multiply by $\frac{1+2i}{1+2i}$: numerator $= (3)(1) - (4)(2) + [(3)(2) + (4)(1)]i = -5 + 10i$; denominator $= 1^2 + 2^2 = 5$. So the quotient is $-1 + 2i$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Powers of i

Evaluate i^{130}, using the fact that powers of i repeat every 4.

a. i **b.** $-i$

c. 1 **d.** -1 ✓

$130 = 4(32) + 2$, so $i^{130} = (i^4)^{32} \cdot i^2 = 1 \cdot -1 = -1$.

22 [Foundational] • Target: Addition and subtraction

Evaluate i^{211}, using the fact that powers of i repeat every 4.

a. 1 **b.** -1

c. $-i$ ✓ **d.** i

$211 = 4(52) + 3$, so $i^{211} = (i^4)^{52} \cdot i^3 = 1 \cdot -i = -i$.

23 [Foundational] • Target: Multiplication

Evaluate i^{229}, using the fact that powers of i repeat every 4.

a. 1 **b.** i ✓

c. -1 **d.** $-i$

$229 = 4(57) + 1$, so $i^{229} = (i^4)^{57} \cdot i^1 = 1 \cdot i = i$.

24 [Foundational] • Target: Conjugate and modulus

Evaluate i^{206}, using the fact that powers of i repeat every 4.

a. i **b.** $-i$

c. -1 ✓ **d.** 1

$206 = 4(51) + 2$, so $i^{206} = (i^4)^{51} \cdot i^2 = 1 \cdot -1 = -1$.

25 [Foundational] • Target: Division

Evaluate i^{62}, using the fact that powers of i repeat every 4.

a. i **b.** -1 ✓

c. 1 **d.** $-i$

$62 = 4(15) + 2$, so $i^{62} = (i^4)^{15} \cdot i^2 = 1 \cdot -1 = -1$.

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26 [Foundational] • Target: Quadratic equations with complex roots

Evaluate i^{101}, using the fact that powers of i repeat every 4.

a. $-i$

b. -1

c. i ✓

d. 1

$101 = 4(25) + 1$, so $i^{101} = (i^4)^{25} \cdot i^1 = 1 \cdot i = i$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate i^{31}, using the fact that powers of i repeat every 4.

a. -1

b. i

c. 1

d. $-i$ ✓

$31 = 4(7) + 3$, so $i^{31} = (i^4)^7 \cdot i^3 = 1 \cdot -i = -i$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate i^{178}, using the fact that powers of i repeat every 4.

a. i

b. 1

c. -1 ✓

d. $-i$

$178 = 4(44) + 2$, so $i^{178} = (i^4)^{44} \cdot i^2 = 1 \cdot -1 = -1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Multiply $(3 - i)(-3 - 9i)$.

a. $-18 - 24i$ ✓

b. $-18 - 30i$

c. $-24i$

d. -18

$(3)(-3) - (-1)(-9) + [(3)(-9) + (-1)(-3)]i = -18 - 24i$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Multiply $(7 + 2i)(4)$.

a. $29 + 8i$

b. $28 + 8i$ ✓

c. $28 - 8i$

d. $28 + 28i$

$(7)(4) - (2)(0) + [(7)(0) + (2)(4)]i = 28 + 8i$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Multiply $(7 - 6i)(4 + 8i)$.

a. $-20 + 32i$

b. $76 + 80i$

c. $76 - 20i$

d. $76 + 32i$ ✓

$(7)(4) - (-6)(8) + [(7)(8) + (-6)(4)]i = 76 + 32i$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Multiply $(-3 - 7i)(1 - 3i)$.

a. $-24 + 18i$

b. $-24 + 2i$ ✓

c. $18 + 2i$

d. $-24 + 16i$

$(-3)(1) - (-7)(-3) + [(-3)(-3) + (-7)(1)]i = -24 + 2i$.

33 [Intermediate] • Target: Powers of i

Multiply $(8 - 5i)(4 + 6i)$.

a. $62 + 2i$

b. $62 + 68i$

c. $62 + 28i$ ✓

d. $2 + 28i$

$(8)(4) - (-5)(6) + [(8)(6) + (-5)(4)]i = 62 + 28i$.

34 [Intermediate] • Target: Addition and subtraction

Multiply $(2 - 2i)(-6 - 5i)$.

a. $-22 - 2i$

b. $-22 + 2i$ ✓

c. $-2 + 2i$

d. $-22 - 22i$

$(2)(-6) - (-2)(-5) + [(2)(-5) + (-2)(-6)]i = -22 + 2i$.

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35 [Intermediate] • Target: Multiplication

Multiply $(-7-6i)(2+3i)$.

a. $4-9i$

b. $4-33i$ ✓

c. $4-32i$

d. $-32-33i$

$(-7)(2)-(-6)(3)+[(-7)(3)+(-6)(2)]i = 4-33i$.

36 [Intermediate] • Target: Conjugate and modulus

Simplify $\frac{6+8i}{2+8i}$ by multiplying top and bottom by the conjugate of the denominator.

a. $-8/17+19/17i$

b. $19/17+8/17i$

c. $19/17-8/17i$ ✓

d. $36/17-8/17i$

Multiply by $(2-8i)/(2-8i)$: numerator $=(6)(2)-(8)(-8)+[(8)(2)+(6)(-8)]i = 76-32i$; denominator $=(2)^2+(8)^2 = 68$. Quotient: $19/17-8/17i$.

37 [Intermediate] • Target: Division

Simplify $\frac{16+4i}{4+2i}$ by multiplying top and bottom by the conjugate of the denominator.

a. $18/5-4/5i$ ✓

b. $-4/5+18/5i$

c. $18/5+4/5i$

d. $23/5-4/5i$

Multiply by $(4-2i)/(4-2i)$: numerator $=(16)(4)-(4)(-2)+[(4)(4)+(16)(-2)]i = 72-16i$; denominator $=(4)^2+(2)^2 = 20$. Quotient: $18/5-4/5i$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 4.01 Advanced Functions & Equations

For $z = 9-40i$, find $|z|$.

a. 49

b. 42

c. 40

d. 41 ✓

$|z| = \sqrt{(9)^2+(-40)^2} = \sqrt{81+1600} = \sqrt{1681} = 41$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

For $z = 5 + 12i$, find $|z|$.

a. 17

b. 12

c. 14

d. 13 ✓

$|z| = \sqrt{(5)^2 + (12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Limits

For $z = 8 - 15i$, find $|z|$.

a. 17 ✓

b. 18

c. 16

d. 23

$|z| = \sqrt{(8)^2 + (-15)^2} = \sqrt{64 + 225} = \sqrt{289} = 17.$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

Solve $x^2 - 14x + 113 = 0$ using the quadratic formula.

a. $7 + 8i$

b. $-7 + 8i,\ -7 - 8i$

c. $8 + 8i, 8 - 8i$

d. $7 + 8i, 7 - 8i$ ✓

Discriminant $= (-14)^2 - 4(113) = 196 - 452 = -256$, so $\sqrt{-256} = 16i$.

$x = (14 \pm 16i)/2 = 7 + 8i, 7 - 8i$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Sequences & Series

Solve $x^2 + 14x + 85 = 0$ using the quadratic formula.

a. $-7 + 6i$

b. $-7 + 6i,\ -7 - 6i$ ✓

c. $-6 + 6i,\ -6 - 6i$

d. $7 + 6i, 7 - 6i$

Discriminant $= (14)^2 - 4(85) = 196 - 340 = -144$, so $\sqrt{-144} = 12i$.

$x = (-14 \pm 12i)/2 = -7 + 6i,\ -7 - 6i$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Functions & Equations

Simplify $\frac{6+8i}{2+8i}$ by multiplying top and bottom by the conjugate of the denominator.

a. $-8/17 + 19/17i$

b. $19/17 + 8/17i$

c. $19/17 - 8/17i$ ✓

d. $36/17 - 8/17i$

Multiply by $(2-8i)/(2-8i)$: numerator $= (6)(2) - (8)(-8) + [(8)(2) + (6)(-8)]i = 76 - 32i$; denominator $= (2)^2 + (8)^2 = 68$. Quotient: $19/17 - 8/17i$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

What is the polar (trigonometric) form of the complex number $z = -2 + 2i\sqrt{3}$?

a. $4(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$ ✓

b. $4(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

c. $2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

d. $4(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

$r = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4+12} = 4$. Angle θ is in Quadrant II: $\theta = \pi - \arctan(\sqrt{3}) = \frac{2\pi}{3}$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Limits

What is the real part of the complex number $(2-2i)^4$?

a. -64 ✓

b. 64

c. 0

d. -32

$2 - 2i = \sqrt{8}e^{-i\pi/4}$. Raising to 4th power: $(\sqrt{8})^4 e^{-i\pi} = 64(-1) = -64$.

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